CineMax Movie Ticket Booking – Whole-System Class Diagram

SE1020 – Object Oriented Programming. One box per workload component (Member 1 to 6) plus the extra modules. Generated from docs/CLASS_DIAGRAM_SYSTEM.md.

M5_BookingSeats
BookingService
+listMine(Long customerId) : List<BookingResponse>
+updateSeats(Long customerId, Long bookingId, BookingUpdateRequest req) : BookingResponse
+cancel(Long customerId, Long bookingId) : CancellationResult
+adminDelete(Long bookingId)
selects by hours before show
notifies
«interface»
CancellationPolicy
+supports(long hoursBeforeShow) : boolean
+refundFor(BigDecimal bookingTotal) : BigDecimal
FullRefundPolicy
PartialRefundPolicy
NoRefundPolicy
manages
makes
Booking
-Long id
-String reference
-Set<String> seatCodes
-BigDecimal totalAmount
-BookingStatus status
+getSeatCount() : int
+markConfirmed()
for show
«enumeration»
BookingStatus
PENDING_PAYMENT
CONFIRMED
CANCELLED
M1_UserAuth
AuthService
+loginCustomer(String email, String password) : User
+loginAdmin(String email, String password) : User
uses
throttles
«interface»
AuthStrategy
+supportedRole() : UserRole
+authenticate(String email, String rawPassword) : User
LoginAttempt
-String emailHash
-int failures
-LocalDateTime lockedUntil
+recordFailure(LocalDateTime now, Duration window, Duration lock, int maxFailures)
CustomerService
+register(RegisterCustomerRequest req) : UserResponse
+updateProfile(Long id, CustomerUpdateRequest req) : UserResponse
+delete(Long id)
«abstract»
User
-Long id
-String fullName
-String email
-String passwordHash
+getRole() : UserRole
AdminService
+createAdmin(CreateAdminRequest req) : UserResponse
+updatePermission(Long actorId, Long id, UpdateAdminRequest req) : UserResponse
+deleteAdmin(Long actorId, Long id)
«abstract»
AbstractAuthStrategy
#findByEmail(String email) : User
manages
manages
Customer
-int loyaltyPoints
+addLoyaltyPoints(int points)
Admin
-String permissionLevel
«enumeration»
UserRole
CUSTOMER
ADMIN
CustomerAuthStrategy
AdminAuthStrategy
1
1
Extra_NotificationsReviewsTicketsReports
TicketService
+ticketPdf(Long customerId, Long bookingId) : byte[]
+qrPng(Long customerId, Long bookingId) : byte[]
reads paid bookings
NotificationService
+inbox(Long customerId) : NotificationInbox
+notifyBookingConfirmed(Booking booking)
+notifyBookingCancelled(Booking booking, BigDecimal refund)
+runReminderSweep()
creates
sends via
Notification
-Long id
-NotificationType type
-String message
-LocalDateTime readAt
+markRead()
«interface»
EmailSender
OutboxEmailSender
SmtpEmailSender
for
written by
ReviewService
+forMovie(Long movieId) : MovieReviews
+create(Long customerId, ReviewRequest request) : ReviewResponse
manages
Review
-Long id
-int rating
-String comment
-Long bookingId
ReportService
+sales(LocalDate from, LocalDate to) : SalesReport
+salesCsv(LocalDate from, LocalDate to) : String
totals, excludes cancelled
M6_PaymentFiles
PaymentService
+pay(Long customerId, PaymentRequest req) : PaymentResponse
+listMine(Long customerId) : List<PaymentResponse>
+updateStatus(Long id, PaymentStatusRequest req) : PaymentResponse
+delete(Long id)
manages
selects by method
Payment
-Long id
-String transactionRef
-BigDecimal amount
-BigDecimal convenienceFee
-BigDecimal refundedAmount
-PaymentStatus status
+refund(BigDecimal refundAmount, BigDecimal bookingTotal)
«interface»
PaymentProcessor
+method() : PaymentMethodType
+validateReference(String reference)
+charge(BigDecimal bookingTotal, String reference) : PaymentCharge
pays for
«enumeration»
PaymentStatus
SUCCESS
REFUNDED
PARTIALLY_REFUNDED
«enumeration»
PaymentMethodType
CARD
WALLET
CASH_COUNTER
«abstract»
AbstractPaymentProcessor
+charge(BigDecimal bookingTotal, String reference) : PaymentCharge
FileService
+importMovies() : ImportResult
+importUsers() : ImportResult
+importShowtimes() : ImportResult
+exportMovies()
+exportShowtimes()
+exportSalesReport() : Path
reads and writes seed file
reads and writes seed file
CardPaymentProcessor
+convenienceFee(BigDecimal bookingTotal) : BigDecimal
WalletPaymentProcessor
CashCounterPaymentProcessor
occupancy
M2_MovieCatalogue
MovieService
+search(String title, String genre) : List<MovieResponse>
+create(MovieRequest req) : MovieResponse
+update(Long id, MovieRequest req) : MovieResponse
+delete(Long id)
about
uses
manages
«factory»
MovieFactory
+create(String format) : Movie
creates
«abstract»
Movie
-Long id
-String title
-String genre
-Integer durationMinutes
-LocalDate releaseDate
+getFormat() : String
+getSurcharge() : BigDecimal
StandardMovie
ImaxMovie
ThreeDMovie
M3_TheatreScreen
TheatreService
+list() : List<TheatreResponse>
+update(Long id, TheatreRequest req) : TheatreResponse
+delete(Long id)
ScreenService
+list(Long theatreId) : List<ScreenResponse>
+create(ScreenRequest req) : ScreenResponse
+update(Long id, ScreenRequest req) : ScreenResponse
+delete(Long id)
manages
uses
Theatre
-Long id
-String name
-String city
«factory»
ScreenFactory
+create(String screenType) : Screen
has
creates
«abstract»
Screen
-Integer rowCount
-Integer seatsPerRow
+getScreenType() : String
+getPriceMultiplier() : BigDecimal
+isValidSeat(String seatCode) : boolean
+getTotalSeats() : int
StandardScreen
PremiumScreen
VipScreen
M4_ShowtimeScheduling
ShowService
+create(ShowRequest req) : ShowResponse
+update(Long id, ShowRequest req) : ShowResponse
+delete(Long id)
manages
Showtime
-Long id
-LocalDateTime startTime
-BigDecimal basePrice
+getMovie() : Movie
priced by highest priority
«interface»
PricingStrategy
+supports(LocalDateTime showTime) : boolean
+priceFor(BigDecimal basePrice, LocalDateTime showTime) : BigDecimal
+priority() : int
WeekdayPricingStrategy
WeekendPricingStrategy
MatineePricingStrategy